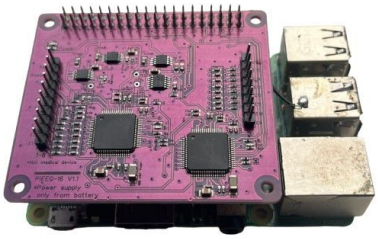
PCB & MANUFACTURING SPECIFICATION · SKU CQA240229G16

Board Material & Fabrication Data

Bare-board material, mechanical features and the production quality test for PiEEG-16 (board revision V1.1). Circuit, layout and stack-up data are proprietary and are not part of this document.



FR-4	1.6 mm	65 × 56.5	Purple	ENIG
Base material	Board thickness	mm, HAT footprint	Solder mask	Recommended finish

01 BARE-BOARD FABRICATION PARAMETERS

Parameters for ordering the bare PCB, in the order used by common PCB order forms. Layer count and stack-up are supplied with the production Gerber package to manufacturing partners only.

Parameter	Specification	Note
Base material	**FR-4**, standard Tg (130–140 °C)	Glass-epoxy laminate
Layers / stack-up	Per production Gerber package	Not published
Dimensions	**65 × 56.5 mm** nominal	Raspberry Pi HAT footprint, rounded corners
Thickness	**1.6 mm**	Standard; matches Pi HAT stand-offs
Solder mask colour	**Purple**	As on production boards
Silkscreen	**White**	Channel numbers, REF, BIAS, safety warnings
Surface finish	**Immersion gold (ENIG)** recommended	Flat pads for two fine-pitch TQFP-64; HASL lead-free acceptable
Copper weight	**1 oz** (35 µm) outer layers	
RoHS	Lead-free materials and process	

02 MECHANICAL FEATURES

Feature	Size	Qty	Purpose
Mounting holes	Ø2.75 mm nominal	4	M2.5 stand-offs, Raspberry Pi HAT pattern 58 × 49 mm
GPIO header	2 × 20, 2.54 mm	1	Stacking header onto the Raspberry Pi
Electrode headers	1 × 10 and 1 × 8, 2.54 mm	2	BIAS, REF, channels 1–8; channels 9–16
Board outline	65 × 56.5 mm	1	Profile per PE16-MD-002

03 ASSEMBLY

Technology	Mixed: surface-mount parts on the top side, through-hole connectors
Key ICs	2 × TI ADS1299 (TQFP-64), with low-noise on-board supplies
Connectors	40-pin extra-high stacking header; 1 × 10 and 1 × 8 electrode headers
Indicators	Green status LEDs
Soldering	Lead-free reflow for SMD, selective or hand soldering for THT
Cleanliness	No-clean flux; keep the analog input areas free of flux residue

For research and education only. Not a medical device. Specifications subject to change.

Marking	“PiEEG-16 V1.1”, “Power supply only from battery”, “Not medical device”

04 PRODUCTION QUALITY TEST

Every assembled board is checked on a Raspberry Pi 5 before shipping, with the PiEEG-16 test script:

1 Power

Mount the shield on a Raspberry Pi running Raspberry Pi OS, power from a battery. **The status LEDs must be on.**

2 Both chips respond

Run the test script. It initialises ADS1299 #1 (CE0) and #2 (GPIO 19) and checks both DRDY lines (GPIO 26 and 13). All 16 channels must show a live trace.

3 Shorted-input check

Set every `CHnSET` on both chips to input-shorted. All 16 channels must stay flat within a few µV.

05 HANDLING, STORAGE AND COMPLIANCE

Topic	Requirement
ESD	Contains ESD-sensitive parts. Handle by the edges on an anti-static surface.
Storage	Dry, in the anti-static bag, 10–35 °C. Keep away from liquids and conductive gel.
Environment	Indoor use only. No conformal coating; do not operate when wet.
Classification	Development and evaluation board for research and education. Not a medical device.
Design files	Schematic, layout, stack-up and BOM are proprietary to PiEEG and are not published.

06 RELATED DOCUMENTS

PE16-DS-001 Product Datasheet	**PE16-MD-002** Mechanical & Dimensional Drawing	**PE16-PIN-004** Connector Pinout & Electrode Guide
PE16-QSG-005 Quick Start & Safety Guide	**PE16-SW-006** Data Interface & Software Guide	**PE16-ENC-007** 3D-Printed Enclosure Guide